INCH-POUND

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20 November 1975

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY 11747210
TARGET COINCIDENCE BOARD

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a target coincidence printed circuit board, hereafter called TCB, a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22, a part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-F-13926 | - Fire Control Materiel, Manufacture and Inspection, General Specification for |
| MIL-I-45607 | - Inspection Equipment, Acquisition, Maintenance and Disposition of |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-810 | - Environmental Test Methods and Engineering Guidelines |
| MIL-STD-2000 | - Standard Requirements for Soldering Electrical and Electronic Assemblies |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

| | |
|---|---|
| F11747212 | - Schematic, Target Coincidence Board |
| F11747210 | - Printed Wiring Board Assembly, Target Coincidence Board |
| F10554938 | - Connector |
| D10548169 | - Integrated Circuit |

PACKAGING DATA SHEET

| | |
|---|---|
| PS11747210 | - Packaging of Target Coincidence Board |

(Copies of drawings, publications and other Government documents required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this documents and the references cited herein the text of this documents takes precedence. Nothing in this documents, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11747212, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Target Coincidence Board (TCB) shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.
e. Soldering electrical connections.

3.3 Environmental service conditions.

3.3.1 Vibration. The TCB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

| Frequency Range | Amplitude |
|---|---|
| 5 to 15 hertz | 0.06 inches double amplitude |
| 15 to 25 hertz | 0.04 inches double amplitude |
| 25 to 55 hertz | 0.02 inches double amplitude |

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The TCB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The TCB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).



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3.3.3.2 Non-operating. The TCB shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The TCB shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32 degrees F and +131 degrees F (equipment used in a sheltered environment), -40 degrees F and +155 degrees F (equipment used in a field environment).

3.3.4.2 Non-operating (storage). The TCB shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The TCB shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the TCB.

3.4.1 Resistance. Resistance shall be 100 ohms ± 10% between the following terminals.

| FROM | TO |
|---|---|
| P1-1 | P1-2 |
| P1-5 | P1-7 |
| P1-20 | P1-22 |

3.5 Performance.

3.5.1 Conditions of operation. Unless otherwise specified, the requirements of 3.5.2 shall be met with the circuit connections shown in Figure 1. Signal input requirements to the circuit of Figure 1 shall comply to the signal requirements given in Table I.



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TABLE I. - Signal inputs.

| ITEM | SIGNAL INPUT | CHARACTERISTICS | APPLICATION POINTS |
|---|---|---|---|
| 1 | Mod $T_O$ | Positive Pulse from 0 V to not less than 15 Vdc. 0.2 to 0.6 usec Pulse Width @ 50% Points. Rise time of 40 nsec max into 50 ohms | Test Fixture J6 |
| 2 | E/L Trigger | Positive Pulse from 0 V to 4.0 ± 0.5 Vdc. Pulse width manually variable between 5 usec and 20 usec. Falling edge occurring at mod $T_O$. PRF=20 KHz ± 200 Hz. | Signal - J1, Terminal 20 Return - J1, Terminal 22 |
| 3 | Target | 14 ± 1 usec positive pulse 0 V to 4.0 ± 0.5 Vdc. Rising edge occurs at Mod $T_O$ | Signal - J1, Terminal 51 Return - J1, Terminal 24 |
| 4 | Doppler | 1 KHz ± 10 Hz Square Wave 0 V to 4.0 ± 0.5 Vdc | Signal - J1, Terminal 35 Return - J1, Terminal 24 |
| 5 | 2560M | 50% Square Wave 0 V to 4.0 ± 0.5 Vdc. Period = 16.38 ± 0.2 Sec. | Signal - J1, Terminal 43 Return - J1, Terminal 24 |

3.5.2 Output signals.

3.5.2.1 Sum video. With the conditions of 3.5.1 applied, the signal present at TP1 on the TCB shall be positive pulse from 0 V to 4.0 ± 1.0 Vdc. The pulse width shall be continuously variable between 0.3 usec and 0.6 usec with variation of R4 located on the TCB. The rising edge of the positive pulse shall be coincident with the falling edge of the target trigger pulse (Item 3, Table I).

3.5.2.2 Difference video. With the conditions of 3.5.1 applied, the signal present at TP2 on the TCB shall be a positive pulse from 0 V to 4.0 ± 1.0 Vdc. The pulse width shall be continously variable between 0.3 usec and 0.6 usec with variations of R5 located on the target coincidence panel. The rising edge of the positive pulse shall be coincident with the falling edge of the sum video pulse (3.5.2.1).



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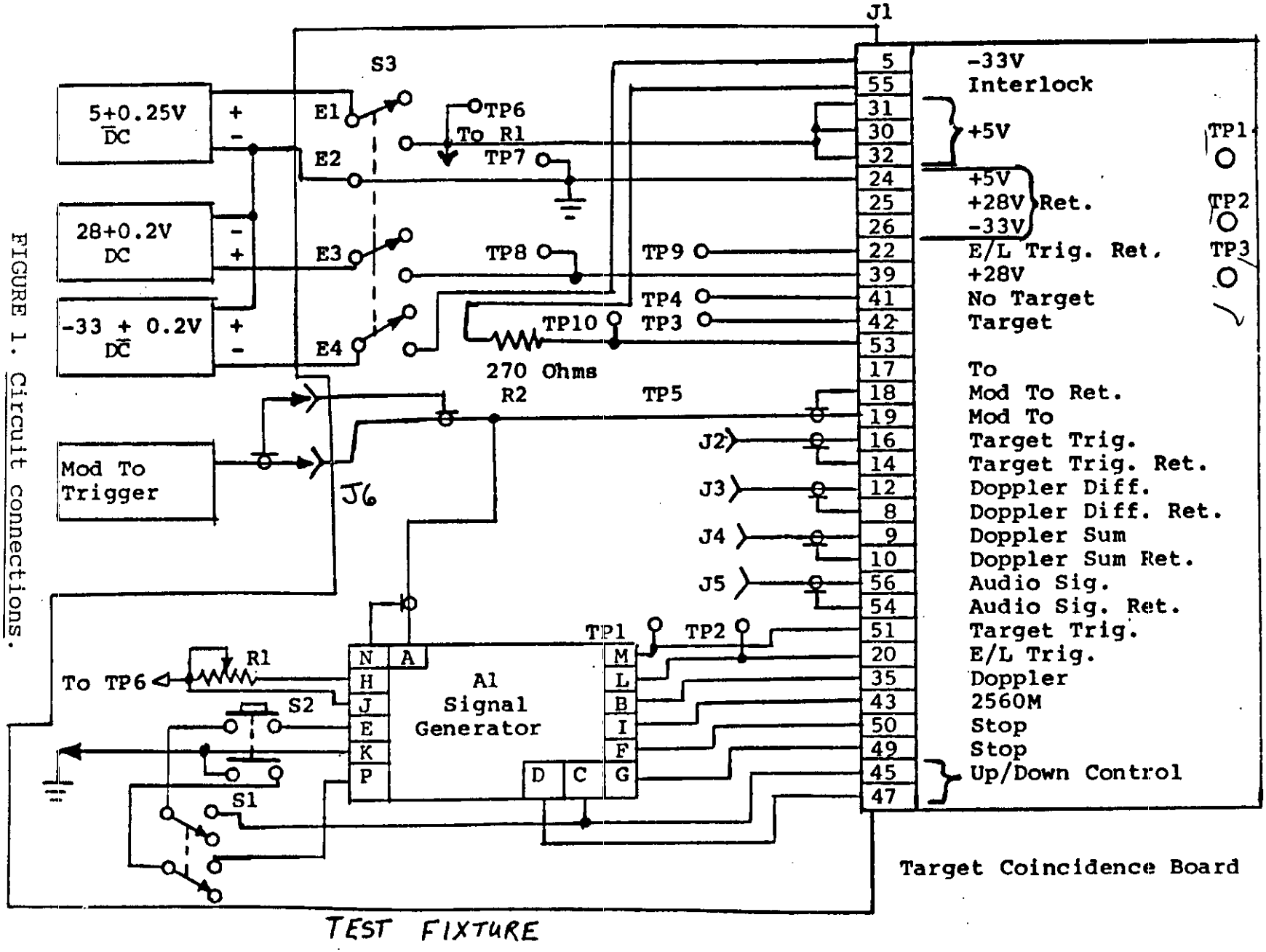



FIGURE 1. Circuit connections.



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3.5.2.3 E/L video. With the conditions of 3.5.1 applied, the signal present at TP3 on the target coincidence board shall be a positive pulse from 0 V to 4.0 ± 1.0 Vdc. The pulse width shall be continuously variable between 0.3 usec and 1.2 usec with variations of R6. The rising edge of the positive pulse shall be coincident with the rising edge of the E/L trigger pulse (Item 2, Table I).

3.5.2.4 Doppler sum. With the conditions of 3.5.1 applied, the signal present at J4 on the test fixture shall be a 1 KHz ± 100 Hz square wave, from 0 V to not less than +0.9 Vdc. This signal shall be present when the E/L video pulse (3.5.2.3) is coincident with either the sum video (3.5.2.1) or difference video (3.5.2.2) pulse. When there is no coincidence, there shall be a dc level of -0.75 ± .01 Vdc at J4.

3.5.2.5 Doppler difference. With the conditions of 3.5.1 applied, the signal present at J3 on the test fixture shall be a 1 KHz ± 100 Hz square wave, from 0 V to not less than +0.9 Vdc. This signal shall be in phase with the DOPPLER SUM (3.5.2.4) when SUM VIDEO only (3.5.2.1) is at least 25 ± 2 nsec coincident with E/L video (3.5.2.3); and 180 degrees out of phase with the SUM DOPPLER (3.5.2.4) when DIFFERENCE VIDEO only (3.5.2.2) is at least 25 ± 2 ns coincident with E/L VIDEO (3.5.2.3). When there is no coincidence, there shall be a dc level of -0.75 ± .01 Vdc.

3.5.2.6 Target Trigger. With the conditions of 3.5.1 applied the signal present on J2 on the test fixture shall be a positive pulse from 0 V to not less than 15 Vdc. The pulse width shall be the same pulse width as the SUM VIDEO (3.5.2.1) pulse, ± 200 ns, and the same PRF.

3.5.2.7 Audio. With the conditions of 3.5.1 applied, the signal present at J5 on the test fixture shall be a 1 KHz ± 10 Hz square wave, 0 V to 4.0 ± 1.0 Vdc. This signal shall be in phase with the DOPPLER (Item 4, Table I) input signal.

3.5.2.8 No target ind. With 4 ± 0.5 Vdc applied to terminal 49 on J1 and 0 V ± 0.5 Vdc applied to terminal 50, the resistance measured between TP3 on test fixture and 24 on J1 shall be less than 2.5 ohms.

3.5.2.9 Target ind. With 0 V, dc applied to terminal 49 on J1 and 4 ± 0.5 Vdc applied to terminal 50, the resistance measured between TP4 on test fixture and 24 on J1 shall be less than 2.5 ohms.



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3.5.2.10 One second fade. When the TARGET TRIGGER (Item 3, Table I) and 2560M (Item 5, Table I) signals are applied, the SUM VIDEO (3.5.2.1) measured on TP1 shall fade for 1 ± 0.2 sec. The fade shall occur 16.4 ± 0.45 sec after 4 ± 0.5 Vdc and 0 Vdc are simultaneously applied to terminal 49 and terminal 50 respectively. The fade shall occur in either position of switch S1.

3.5.2.11 Mod $T_O$. With the Mod $T_O$ trigger (Item 1, Table 1) applied, the signal present at TP5 on the test fixture shall be a positive pulse from 0.0 Vdc to 4.0 ± 1.0 Vdc, 0.6 to 1.2 usec pulse width and a PRF of 20.0 ± 0.2 Khz.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5) (if required).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

| | | |
|---|---|---|
| a. | Temperature | +50 deg to +90 deg F. |
| b. | Relative humidity | 90% RH Max. |
| c. | Pressure | 28 to 32 inches of Hg. |

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of three manufactured in the same manner, using the same materials, equipment, processes, and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including Certified Test Report (CTR) or Certificate Of Conformance (COC) for the materials.

4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table II (Functional test) and Table III (Special sampling) herein and to all requirements of the applicable drawings. The sample and a record of this inspection, including CTR/COC, shall be submitted for the Government approval. The Government reserves the right to witness the contractor's inspection.



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4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine comformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.



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TABLE II. Functional tests.
(100% Inspection)

| | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| Critical: | | None | N/A |
| Major: | | | |
| 101. | Electrical | 3.4 | 4.7.2 |
| 102. | Performance | 3.5 | 4.7.3 |
| Minor: | | None | N/A |

4.5 Special sampling. (Note: Employ when environmental testing is needed). One TCB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling

| | Characteristic | Requirements | Test Procedure |
|---|---|---|---|
| Critical: | | None | N/A |
| Major: | | | |
| 103. | Vibration | 3.3.1 | 4.7.1.1 |
| 104. | Shock | 3.3.2 | 4.7.1.2 |
| 105. | Amplitude | 3.3.3 | 4.7.1.3 |
| 106. | Temperature | 3.3.4 | 4.7.1.4 |
| 107. | Humidity | 3.3.5 | 4.7.1.5 |
| Minor: | | None | N/A |

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.



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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.



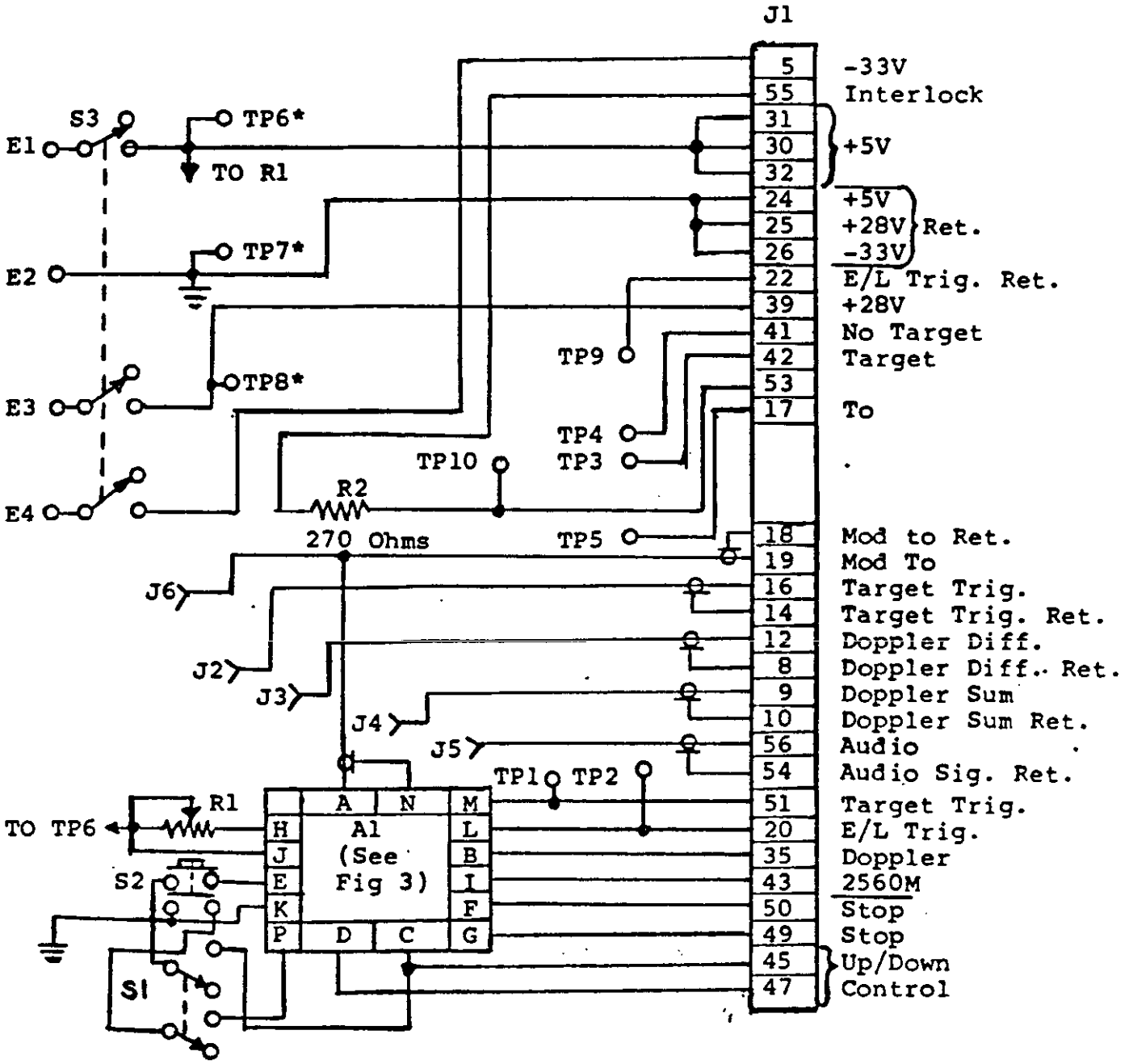
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TABLE IV. - Test equipment list.

| ITEM | EQUIPMENT | DESCRIPTION |
|---|---|---|
| 1. | Power supply | 5.0 ± 0.25 Vdc, Regulated AC ripple: Less than 50 MV peak-to-peak Capacity: 600 ma. |
| 2. | Power supply | 28 ± 0.2 Vdc, Regulated AC ripple: Less than 50 mv peak-to-peak Capacity: 200 ma. |
| 3. | Signal generator | 300 ± 10 ns wide pulse, repetition rate 20 KHz ± 100 Hz, 10 ± 0.1V peak-to-peak into 50 ohms. |
| 4. | Oscilloscope | Dual channel; rise time of scope plus probe not greater than 10 nsec, sweep period less than 50 nsec Input resistance: 8 megohms or greater. |
| 5. | Test fixture | As specified in Figure 2. See Table V for test fixture parts list. |
| 6. | Ohmmeter | Accuracy and resolution of ±10%, to measure 200 ohms. |
| 7. | Power supply | 0 to +35V adj., regulated AC ripple: Less than 50 mv peak-to-peak Capacity: 400 ma. |
| 8. | Voltmeter | Accuracy and resolution of ±1%, 0 to 10 Volts, Digital. |



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*Connect TP6 thru TP8 as close to J1 as possible.

FIGURE 2. Test fixture.



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TABLE V. Test fixture parts list.

| DESIGNATION | NOMENCLATURE AND DESCRIPTION |
|---|---|
| A1 | Test fixture signal generator in accordance with Figure 3. See Table VI for parts list. |
| E1, E3, E4 | Binding post, H. H. Smith 257-R or equivalent. |
| E2 | Binding post, H. H. Smith 257-R or equivalent. |
| J1 | Connector, per Frankford Arsenal F10554938 or equivalent. |
| J2 thru J6 | Connector, BNC Receptacle |
| TP1 thru TP 10 | Tip Jack, H. H. Smith 1501 or equivalent. |
| S1 | Switch DPDT. |
| S2 | Switch DPST. |
| S3 | Switch 3 PDT. |
| R1 | Variable Resistor, 10K Ohms, 1/4 watt, 10 turn. |
| R2 | Resistor, 270 Ohms $\pm$ 1%, 5 watt. |



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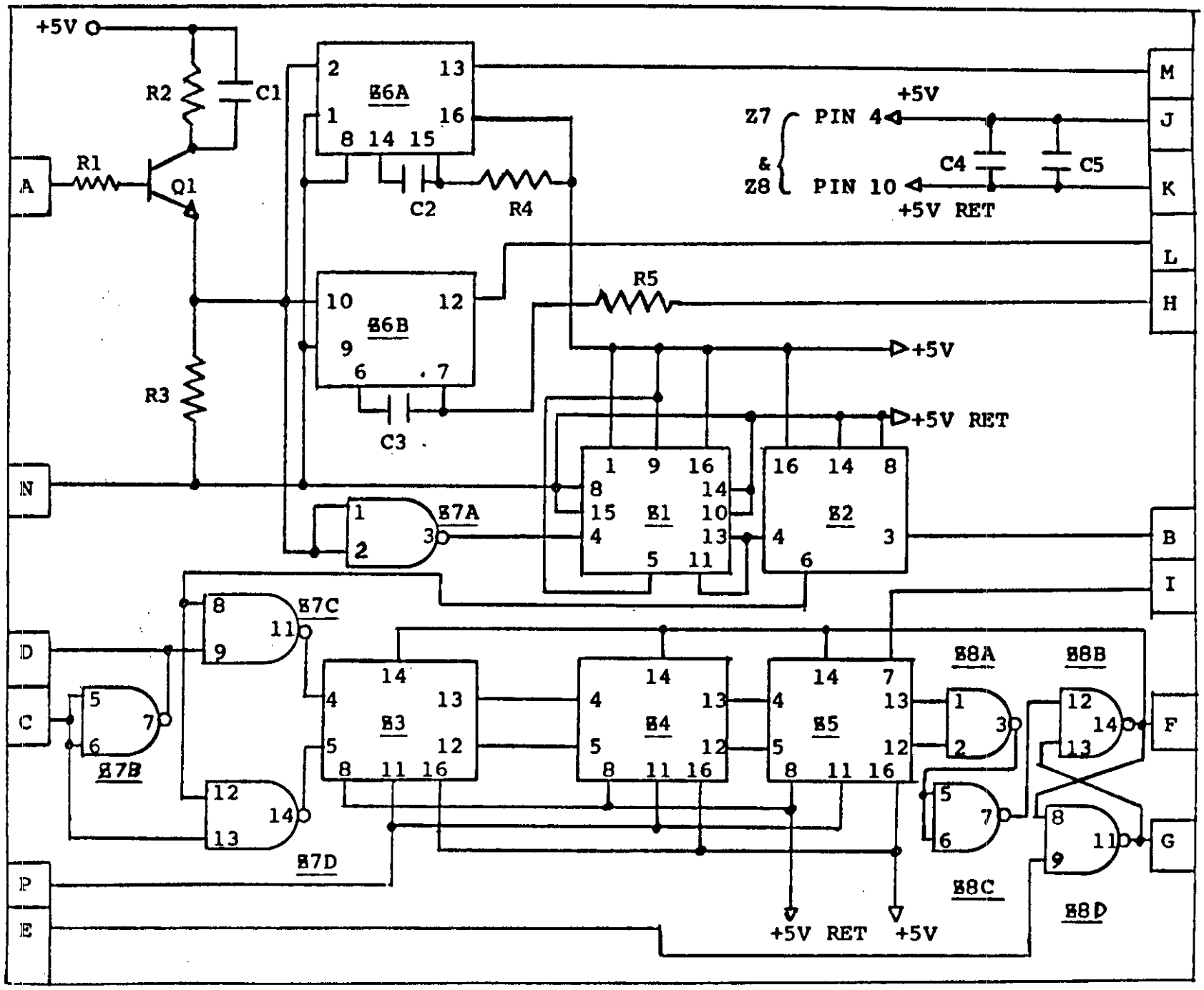



FIGURE 3. Test fixture signal generator A1.



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TABLE VI. Test fixture signal generator parts list.

| DESIGNATION | NOMENCLATURE AND DESCRIPTION |
|---|---|
| C1, C4, C5 | Capacitor 0.1 uF ± 10%, @ 50 Vdc |
| C2, C3 | Capacitor 4700 pF ± 10%, @ 50 Vdc |
| Q1 | Transistor 2N222 |
| R1, R4 | Resistor, 10K ohms ± 10%, 1/4 watt |
| R2, R3 | Resistor, 100 ohms ± 10%, 1/4 watt |
| Z1, Z2, Z3, Z4, Z5 | Integrated Circuit Texas Instruments, P/N SN54193 |
| Z6 | Integrated Circuit Texas Instruments, P/N SN54123J |
| Z7, Z8 | Integrated Circuit FA, P/N 10548169 |
| R5 | Resistor, 3K ohms ± 10%, 1/4 watt |

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph II-2.1 apply. The TCB shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the TCB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.



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d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the TCB shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2. Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical.

4.7.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.7.2.2 Resistance. Resistance shall be measured with an ohmmeter (item 6 of Table IV) for compliance with 3.4.2.

4.7.3 Performance.

4.7.3.1 Conditions of Operation. Unless otherwise specified, all subsequent tests will be performed with the DC power supply connections and signal connections as follows: Connect up the power supplies (items 1, 2, and 7 of Table IV), the signal generator (item 3 of Table IV), and the test fixture, (item 5 of Table IV) to conform to the schematic of Figure 1. Adjust the DC power supplies to the values indicated. Apply the MOD $T_O$ trigger (item 1, Table I) to test fixture J6.



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4.7.3.2 Sum Video. Connect channel 1 of the oscilloscope, item 4 of Table IV, to TP1 on the TCB. Check for the presence of the positive pulse per 3.5.2.1. Vary R4 on the TCB and check the pulse width for conformance to 3.5.2.1 requirements. Connect channel 2 of the oscilloscope to TP1 of the test fixture (+) and E2 of the test fixture (-), and check the phase requirements of 3.5.2.1.

4.7.3.3 Difference Video. Connect channel 1 of the oscilloscope, item 4 of Table IV, to TP2 on the TCB. Check for the presence of the positive pulse per 3.5.2.2. Vary R5 on the TCB and check the pulse width for conformance to 3.5.2.2 requirements. Connect channel 2 of the oscilloscope to TP1 on the TCB, and eck the phase requirements of 3.5.2.2.

4.7.3.4 E/L Video. Connect channel 1 of the oscilloscope, item 4 of Table IV, to TP3 on the TCB. Check for the presence of the positive pulse per 3.5.2.3. Vary R6 on the TCB and check the pulse width for conformance to 3.5.2.3. Connect channel 2 of the oscilloscope to TP2 of the test fixture (+) and TP9 of the test fixture (-), and check for the phase requirements of 3.5.2.3.

4.7.3.5 Doppler Sum. Connect channel 1 of the oscilloscope, item 4 of Table IV, to J4 of the test fixture. Check for the presence of the 1 KHz square wave defined in 3.5.2.4. If the square wave is present, disconnect the oscilloscope. Connect channel 1 of the oscilloscope to TP3 on the TCB, and channel 2 alternately to TP1 and TP2. Check for coincidence of the E/L video pulse, with either the SUM video or difference video pulses per the requirements of 3.5.2.4. If the square wave is not present at terminal J4 check for the dc level per 3.5.2.4 requirements.

4.7.3.6 Doppler difference. Connect channel 1 of the oscilloscope, item 4 of Table IV, to J3 on the test fixture. Check for the presence of the 1 KHz square wave per the requirements of 3.5.2.5.

a. If the signal is not present, go to b. If the signal is present, connect channel 2 of the oscilloscope to J4 of the test fixture. Check to see if channel 2 signal is in-phase or out-of-phase with channel 1 signal. If signals are in phase, connect channel 1 to TP1 and channel 2 to TP3. Determine that the two displayed signals are at least 25 $\pm$ 2 ns coincident per the requirements of 3.5.2.5. If signals are out-of-phase, connect channel 1 to TP2 and channel 2 to TP3. Determine that the two displayed signals are at least 25 $\pm$ 2 ns coincident per the requirements of 3.5.2.5.

b. When the signal is not present, check for a dc level of -0.86 $\pm$ .04 per the requirement of 3.5.2.5.



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4.7.3.7 Target trigger. Connect channel 1 of the oscilloscope, item 4 of Table IV, to J2 on the test fixture. Check for the presence of the positive pulse per the requirements of 3.5.2.6. Connect channel 2 of the oscilloscope to TP1 on the TCB. Check for pulse width and PRF per the requirements of 3.5.2.6.

4.7.3.8 Audio. Connect channel 1 of the oscilloscope, item 4 of Table IV, to J5 on the test fixture. Check for the presence of the 1 KHz square wave per the requirements of 3.5.2.7. Connect channel 2 to the application points of the Doppler signal given in Table I, as item 4. Check for the phase requirements of 3.5.2.7.

4.7.3.9 No target ind. Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 4 $\pm$ 0.5 Vdc per the requirements of 3.5.2.8. If the dc signal is not present, momentarily depress switch S2. When the dc signal is present, proceed to measure the resistance between TP3 on test fixture and 24 of J1 using an ohmmeter, item 6 of Table IV. Determine that the resistance measured meets the requirements of 3.5.2.8.

4.7.3.10 Target ind. Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 0 $\pm$ 0.2 Vdc per the requirements of 3.5.2.9. If the dc signal is not present, momentarily depress switch S2. When the dc signal is present, proceed to measure the resistance between TP4 on test fixture and 24 of J1 using an ohmmeter, item 6 of Table IV. Determine that the resistance measured meets the requirements of 3.5.2.9.

4.7.3.11 One second fade. The steps given below shall be followed in sequence.

a. Connect a voltmeter, item 8 of Table IV, to terminal 49 of J1(+) and return to terminal 50(-). Check for the presence of 0 $\pm$ 0.2 Vdc. If the dc signal is not present, momentarily depress switch S2 on the TCB. When the dc signal is present, disconnect the voltmeter.

b. Check for the presence of the TARGET TRIGGER (item 3, Table I) signal, and the 2560M (item 5, Table I) signal, by using the oscilloscope, item 4 of Table IV, on the application points for these signals listed in Table I.

c. Connect channel 1 of the oscilloscope to TP1 on the TCB.

d. Monitor signal on oscilloscope and depress switch S2 momentarily. Confirm the presence of signal fade per the requirements of 3.5.2.10. Change the position of S1 on the test fixture and repeat steps a through d.



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4.7.3.12 MOD $T_O$. Connect channel 1 of the oscilloscope, item 4 of Table IV, to TP5 on the test fixture. With the conditions as specified in 3.5.2.11 check for the presence of the pulse as specified in 3.5.2.11.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 Packaging. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11747210. The level of protection shall be as specified in the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This Circuit Card Assembly is a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22 which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.



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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Range Computer Test Panel
Radar Test Set, AN/TPM-22
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A293)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-C-48581A (AR) | 2 February 1991 |

3. DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY 11747210 TARGET COINCIDENCE BOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

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|---|---|---|
| US ARMY ARDEC<br>STANDARDIZATION OFFICE | (1) Commercial<br>(201) 724-6662 | (2) AUTOVON<br>880-6662 |
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